RESUME

Jared Lemler

Backend Engineer · Pacifica, CA

jaredlemler.com | github.com/JL037 | linkedin.com/in/jaredlemler

PROFESSIONAL SUMMARY

Backend Engineer with a craftsman's approach — 8 years in electrical systems, now building software that is safe, reliable, and maintainable. My electrician background honed my systems thinking: coordinating teams, reading complex plans, and building infrastructure that can't fail. I bring the same patience and troubleshooting discipline from wiring buildings to designing APIs, databases, and containerized services.

Technically grounded in Python/FastAPI and TypeScript/Fastify, with production experience in PostgreSQL, Prisma, Docker. Recently architected VoxPort, a federated podcast platform on the AT Protocol, building high-throughput data pipelines (1,000 event bounded in-memory queue, 100 event batched persistence, ~95% I/O reduction) under real-world constraints. I work with honest communication, clear ownership, and passion for contributing to teams.

PROJECTS

VoxPort voxport.live → FASTIFY · NEXT.JS · ATPROTO

Podcast Platform built on the AT Protocol

- Zero-Downtime AT Protocol Federation: Built resilient WebSocket consumer surviving 10 reconnection cycles with exponential backoff and cooloff, with bounded memory backpressure (1,000 event cap per worker), processing 6 collections with 100 event batching. Achieved ~95% I/O reduction vs. naive per-event persistence.
- Engagement Analytics Engine: Built hourly velocity aggregation system tracking engagement across 1h to 168h (7-day) windows using a weighted scoring algorithm (likes=1x, mentions=5x) for real-time trending detection via hourly scheduled jobs.
- Memory Optimized Cursor Tracking: Eliminated per-event DB writes via in-memory cursor with 5-second flush intervals, reducing database round trips by ~95% during high-volume ingestion.
- Relational Data Architecture: Designed 26 entity normalized schema with time-series aggregation tables for historical trend analysis.
- Scalable Media Infrastructure: Integrated Cloudflare R2 for content-addressed blob storage, mapping CIDs to AT Protocol records for verifiable media distribution across federated nodes.
- Backfill Migration System: Developed ETL pipeline for migrating legacy content to AT Protocol, supporting bulk series/episode migration with dry-run validation and error tracking.
- Zero-Trust Audio Security: Implemented magic-number file validation (not extensions), ffmpeg metadata stripping, and 450MB limits — blocking spoofed uploads while preserving creator metadata privacy.
- Distributed Extraction Worker: Built standalone Fly.io worker claiming MediaJob rows via `FOR UPDATE SKIP LOCKED` with per-source-host politeness gating (1 concurrent job per host), exponential backoff retry (30s/60s/120s, max 3 attempts), graceful shutdown finishing in-flight jobs, and scale-to-zero idle behavior. Keeping ffmpeg and outbound fetches out of request-serving API.

doc-lok npm → github → TYPESCRIPT · NODE.JS · ZERO-DEP

Pre-prompt context condenser

- Context Window Optimization: Engineered a TypeScript pre-prompt middleware that condenses prompt payloads by resolving remote markdown links into compact cryptographic markers (<lok_id>), drastically cutting context token consumption and API latency.
- Conditional Caching Pipeline: Built a two-phase network verification engine using HTTP HEAD probes and If-None-Match ETag validation to bypass full payload downloads for unchanged remote resources.
- Zero-Dependency Architecture: Authored a lightweight CLI and library relying exclusively on native Node.js APIs, eliminating third-party supply chain vulnerabilities and minimizing cold-start overhead.

Travel Wishlist travelwishlist.app → FASTAPI · REACT · DOCKER · POSTGRES

Location-Based Travel Planner

- Architecture & Design: Built with FastAPI, PostgreSQL, and Docker. Implemented JWT authentication, Pydantic validation, and a reusable CRUD service layer.
- Weather API Integration: Built city-based weather lookup system integrating external weather APIs with 100ms response times.
- Calendar-Based Itinerary System: Created monthly/agenda calendar views for trip planning with date-based organization of saved cities and travel scheduling.
- Interactive Map Feature: Mapping layer for geographic visualization of saved destinations.

SKILLS

LANGUAGES
Python, JavaScript, TypeScript, SQL

DATABASES
PostgreSQL, Prisma Postgres, Neon, Alembic migrations, schema design, indexing, time-series aggregation

DATA & ETL
Pandas, Pydantic Validators, Scheduled Jobs, batch processing, CSV Ingestion, JSON Pipelines

DEVOPS & TOOLING
Docker, Alembic, Git, GitHub, JWT, pytest, Node.js, Sentry, OpenTelemetry, structured logging, Axiom

CLOUD & INFRASTRUCTURE
AWS (EC2, Elastic Beanstalk, S3), CI/CD (CodeQL, Gitleaks, Bandit), Cloudflare R2, Hetzner, Neon, Fly.io

FRAMEWORKS
FastAPI, Fastify, SQLAlchemy 2.0, Pydantic, Django, React, Prisma ORM

EXPERIENCE

Backend Developer (Self-Employed)
Jan 2025 – Feb 2026

Talk — Social Connection Platform (Netherlands)

- Developed and deployed a privacy-compliant web app supporting 200+ university students using React, Vite, Tailwind, and FastAPI backend integrations.

- Translated founder vision into technical implementation using Lovable prototypes, delivering full-stack solutions.
- Created a QR code to trigger data entries between physical and digital experiences for social experiments, enabling real-time participation and improving data collection efficiency.
- Built automated data pipelines and dashboards converting raw activity into insights, reducing manual reporting effort by 90%.
- Consulted with team and users, leading to decreased onboarding steps and increased user retention in live trials.
- Collaborated with founders to align analytics and engagement metrics for research and product direction.

TypeScript · React · Tailwind · Vite · Cloudflare · Plausible Analytics

Journeyman Electrician

2017 – 2025

Gaylor Electric · Wolverine Electric · IBEW — Indiana & Michigan

- Coordinated crews of 2–4 electricians on commercial jobsites, reading and interpreting complex electrical and architectural plans to execute installations on schedule and to code.
- Built and maintained electrical infrastructure with zero tolerance for failure — troubleshooting live systems under time pressure and safety constraints, the same discipline I now apply to APIs, databases, and containerized services.
- Completed electrical training through Associated Builders and Contractors (ABC) trade school and earned journeyman status; remain a dues-paying IBEW member able to return to union dispatch.

CORE COMPETENCIES

REST API Design · Authentication & Authorization · Data Modeling · Distributed Systems · AT Protocol · WebSocket · Real-time Event Processing

EDUCATION & CERTIFICATIONS

Journeyman Electrician — Associated Builders and Contractors (ABC) Indiana

Completed electrical trade training through ABC, combining classroom instruction with on-the-job experience; earned journeyman status. Dues-paying IBEW member.

Self-directed Software Engineering Study Ongoing

Python, FastAPI, PostgreSQL, distributed systems, REST API design, and AT Protocol — learned through building and shipping production software (VoxPort, doc-lok, Travel Wishlist) and through ongoing mentorship from senior engineers, including former Facebook and Microsoft engineers and a current Prisma platform architect.